## Topic to disscuss

- → Bessel's central difference Interpolation
- → Numerical Problem
- → Homework Problem with solution.

# Bessel's Interpolation Formula

The formula is :

$$y(x) = \frac{y_0 + y_1}{2} + \left(u - \frac{1}{2}\right) \Delta y_0 + \frac{u(u-1)}{2!} \cdot \left[\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2}\right] + \frac{\left(u - \frac{1}{2}\right) u (u-1)}{3!} \Delta^3 y_{-1}$$

$$+ \frac{(u+1) u (u-1)(u-2)}{4!} \left[\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2}\right] + \cdots$$

where $u = \frac{x - x_0}{h}$

and $\frac{1}{4} \leq u \leq \frac{3}{4}$

Q:- Using Bessel's Interpolation, solve this and find the value of $y$ at $x = 11.7$

| $x$ | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|
| $y$ | 3.16 | 3.32 | 3.46 | 3.61 | 3.74 |

Solution: Step length : $h = 11 - 10 = 1$

Choosen central value : $x_0 = 11$

Interpolation point : $x = 11.7$

$$u = \frac{x - x_0}{h} = \frac{11.7 - 11}{1}$$

$$\therefore u = 0.7$$

# Bessels Central difference Table

| | $x$ | $y$ | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ | $\Delta^4 y$ |
|---|---|---|---|---|---|---|
| $x_{-1}$ | 10 | $y_{-1} = 3.16$ | $\Delta y_{-1} = 0.16$ | $\Delta^2 y_{-1} = -0.02$ | $\Delta^3 y_{-1} = 0.03$ | $\Delta^4 y_{-1} = -0.06$ |
| $x_0$ | 11 | $y_0 = 3.32$ | $\Delta y_0 = 0.14$ | $\Delta^2 y_0 = 0.01$ | $\Delta^3 y_0 = -0.03$ | |
| $x_1$ | 12 | $y_1 = 3.46$ | $\Delta y_1 = 0.15$ | $\Delta^2 y_1 = -0.02$ | | |
| $x_2$ | 13 | $y_2 = 3.61$ | $\Delta y_2 = 0.13$ | | | |
| $x_3$ | 14 | $y_3 = 3.74$ | | | | |

Bessels central difference formula,

$$y(x) = \frac{y_0 + y_1}{2} + \left(u - \frac{1}{2}\right)\Delta y_0 + \frac{u(u-1)}{2!}\cdot\left[\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2}\right] + \frac{\left(u-\frac{1}{2}\right)u(u-1)}{3!}\Delta^3 y_{-1}$$

$$+ \frac{(u+1)u(u-1)(u-2)}{4!}\left[\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2}\right]$$

$$= \frac{3.32 + 3.46}{2} + \left(0.7 - \frac{1}{2}\right)\times 0.14 + \frac{0.7(0.7-1)}{2!}\left(\frac{-0.02 + 0.01}{2}\right)$$

$$+ \frac{\left(0.7 - \frac{1}{2}\right)0.7\times(0.7-1)}{3!}\times 0.03$$

$$y(x) = 3.39 + 0.028 + 0.000525 - 0.00021$$

$$= 3.418315$$ Ans.

## Homework problem :

Using Bessel's Interpolation formula, find $y$ at $x = 26.6$ from the table :

| $x$ | 24 | 25 | 26 | 27 | 88 |
|---|---|---|---|---|---|
| $y$ | 4.899 | 5.000 | 5.099 | 5.196 | 5.292 |

solution : Step length : $h = 25 - 24 = 1$

Choosen central value : $x_0 = 26$

Interpolation point : $x = 26.6$

$$\therefore u = \frac{x - x_0}{h} = \frac{26.6 - 26}{1}$$

$$\therefore u = 0.6$$

Bessel's Central difference table :

| | $x$ | $y$ | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ | $\Delta^4 y$ |
|---|---|---|---|---|---|---|
| $x_{-2}$ | 24 | 4.899 | | | | |
| | | | 0.1010 | | | |
| | | | | -0.002 | | |
| $x_{-1}$ | 25 | 5.000 | | | 0 | |
| | | | 0.099 | | | 0.001 |
| | | | | -0.002 | | |
| $x_0$ | 26 | 5.099 | | | 0.001 | |
| | | | 0.097 | | | |
| | | | | -0.001 | | |
| $x_1$ | 27 | 5.196 | | | | |
| | | | 0.096 | | | |
| $x_2$ | 28 | 5.292 | | | | |

Bessels central difference formula,

$$y(x) = \frac{y_0 + y_1}{2} + \left(u - \frac{1}{2}\right)\Delta y_0 + \frac{u(u-1)}{2!}\cdot\left[\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2}\right] + \frac{\left(u-\frac{1}{2}\right)u(u-1)}{3!}\Delta^3 y_{-1}$$

$$+ \frac{(u+1)u(u-1)(u-2)}{4!}\left[\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2}\right]$$

$$= \frac{5.099 + 5.196}{2} + \left(0.6 - \frac{1}{2}\right) \times 0.097 + \frac{0.6(0.6-1)}{2!}\left(\frac{-0.002 - 0.001}{2}\right)$$

$$+ \frac{\left(0.6 - \frac{1}{2}\right)0.6(0.6-1)}{3!} \times 0.001$$

$$= 5.1475 + 0.0097 + 0.00018 - 0.000004$$

$$= 5.157376$$ Ans.

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